The cvd package (v0.1.0)

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1 Introduction

2 Implementation

2.1 Package Dependencies

Load required packages.

```
\RequirePackage{iftex}
\RequirePackage{xcolor}
% for optional \includegraphics patching
\RequirePackage{keyval}
```

2.2 Engine Check

Currently only LuaTeX is fully supported.

```
\ifluatex\else
  \PackageError{cvd}{LuaTeX required}{%
    This package currently only works with LuaLaTeX.\MessageBreak
    pdfLaTeX support is under development.}
\fi
```

2.3 Color Space Enforcement

Force RGB color model for consistent transformations.

```
\selectcolormodel{rgb}
```

2.4 Load Lua Module

Next, load the Lua module that implements the CVD transformations. The `install_pdf_image_hook` function registers a callback that transforms colors in embedded PDF pages (vector graphics only). We also load the Lua File System module for file timestamp checking.

```
\directlua{lfs = require("lfs"); cvd = require("cvd"); cvd.install_pdf_image_hook()}
```

2.5 Hook into xcolor

Use `xcolor`'s `\XC@bcolor` hook to transform RGB values before display. This handles text colors, color boxes, and other `xcolor`-based content.

```
\def\XC@bcolor{%
  \directlua{token.set_macro("current@color", cvd.transform_current_color("\luaescapestring{\c
}
```

2.6 User Commands

`\cvdtype` Set the type of color vision deficiency to simulate.

```
\newcommand{\cvdtype}[1]{%
  \directlua{cvd.set_type("#1")}%
}
```

`\cvdseverity` Set the severity of the simulation (0.0 to 1.0).

```
\newcommand{\cvdseverity}[1]{%
  \directlua{cvd.set_severity(#1)}%
}
```

`\cvdalgorithm` Set the algorithm to use (`brettel` or `machado`).

```
\newcommand{\cvdalgorithm}[1]{%
  \directlua{cvd.set_algorithm("#1")}%
}
```

`\cvdenable` Enable CVD simulation.

```
\newcommand{\cvdenable}{%
  \directlua{cvd.enable()}%
}
```

`\cvddisable` Disable CVD simulation.

```
\newcommand{\cvddisable}{%
  \directlua{cvd.disable()}%
}
```

`\cvdincludegraphics` Include a graphics file with CVD transformation applied to raster images.

```
\newcommand{\cvdincludegraphics}[2][]{%
  \edef\cvd@imgpath{\directlua{tex.sprint(cvd.get_image_path("\luaescapestring{#2}"))}}%
  \includegraphics[#1]{\cvd@imgpath}%
}
```

2.7 Package Options

Handle package options for deficiency type and severity.

```
\DeclareOption{protanopia}{\AtEndOfPackage{\cvdtype{protanopia}\cvdseverity{1.0}}}
\DeclareOption{deuteranopia}{\AtEndOfPackage{\cvdtype{deuteranopia}\cvdseverity{1.0}}}
\DeclareOption{tritanopia}{\AtEndOfPackage{\cvdtype{tritanopia}\cvdseverity{1.0}}}
\DeclareOption{protanomaly}{\AtEndOfPackage{\cvdtype{protanopia}\cvdseverity{0.5}}}
\DeclareOption{deuteranomaly}{\AtEndOfPackage{\cvdtype{deuteranopia}\cvdseverity{0.5}}}
\DeclareOption{tritanomaly}{\AtEndOfPackage{\cvdtype{tritanopia}\cvdseverity{0.5}}}
\DeclareOption{brettel}{\AtEndOfPackage{\cvdalgorithm{brettel}}}
\DeclareOption{machado}{\AtEndOfPackage{\cvdalgorithm{machado}}}
\DeclareOption*{\PackageWarning{cvd}{Unknown option `\CurrentOption'}}
\ProcessOptions\relax
```